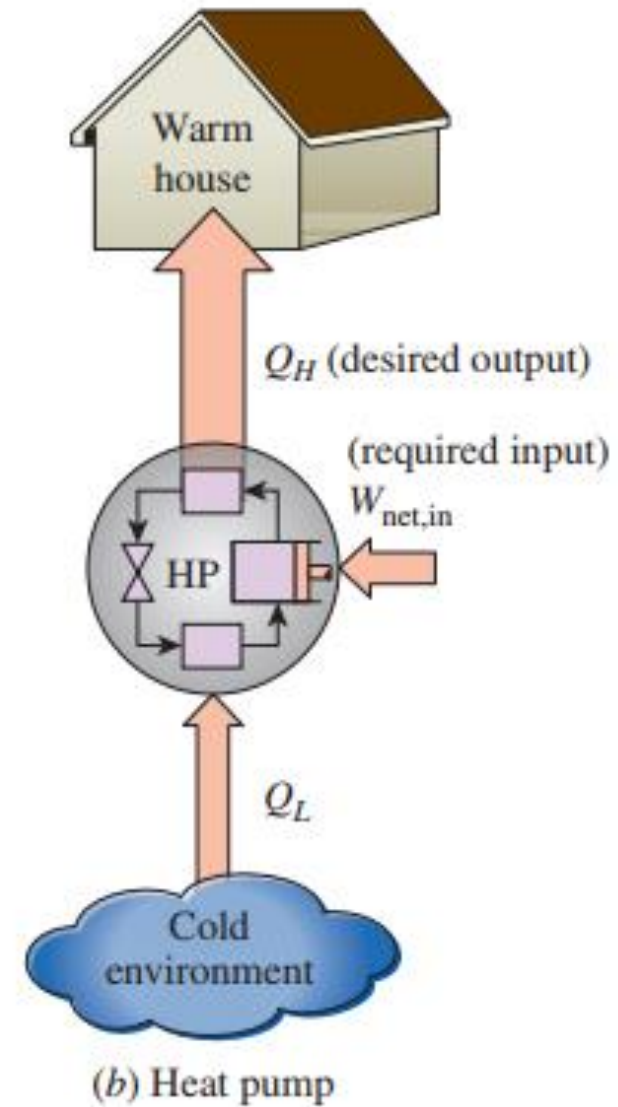
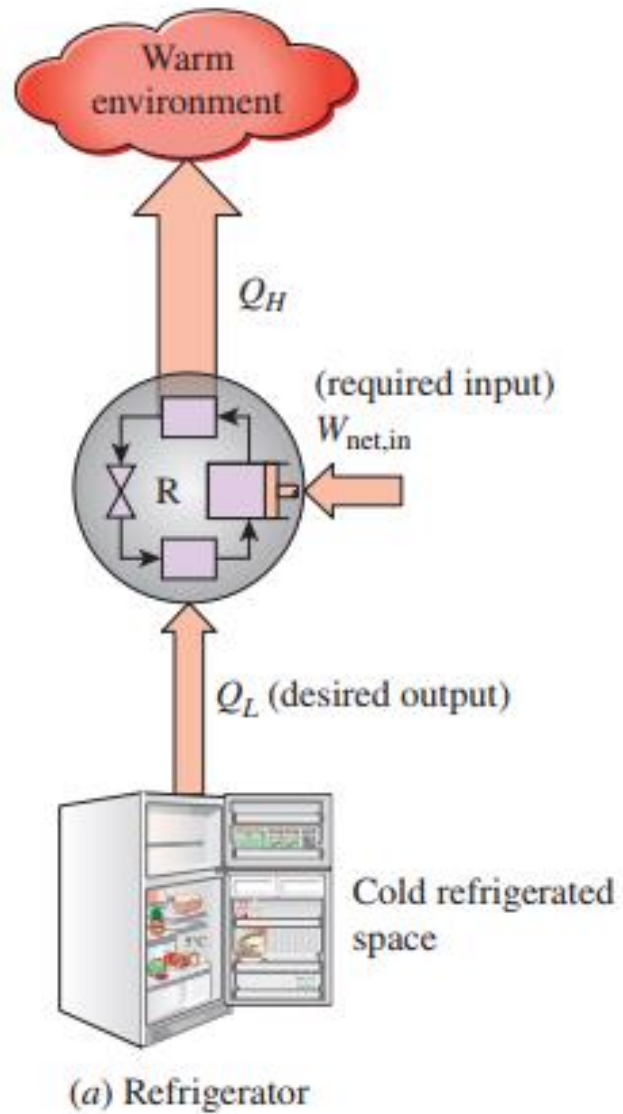


Chapter 3 Refrigeration and Heat Pump System

Introduction

- The transfer of heat from a low-temperature region to a high-temperature one requires special devices called **refrigerators**.
- Refrigerators are cyclic devices, and the working fluids used in the refrigeration cycles are called **refrigerants**.
- Another device that transfers heat from a low-temperature medium to a high temperature one is the **heat pump**.
Refrigerators and heat pumps are essentially the same devices; they differ in their objectives only.
- **Refrigerators** put work in to cool a space using Q_{in} (cooling load)
- **Heat Pump** put work in to heat a space using Q_{out} (heating load)



The objective of a refrigerator is to remove heat (Q_L) from the cold medium; the objective of a heat pump is to supply heat (Q_H) to a warm medium.

$Q_H = W_{in} + Q_L$, dividing the whole by W_{in} , we get

$$\frac{Q_H}{W_{net,in}} = \frac{W_{net,in}}{W_{net,in}} + \frac{Q_L}{W_{net,in}}$$

$COP_{HP} = COP_R + 1$ for fixed values Q_L and Q_H

Units of Refrigeration

- The cooling capacity of a refrigeration system—that is, the rate of heat removal from the refrigerated space—is often expressed in terms of **tons of refrigeration**.
- In commercial practice, the specification of the cooling rate is expressed in tons of refrigeration.
- The capacity of a refrigeration system that can freeze 1 ton of liquid water at 0°C into ice at 0°C in 24 h is said to be 1 ton.
- A “ton” of refrigeration corresponds to the cooling (freezing) of 1 ton(907.19 kg) of ice at 0°C per day.

The performance of refrigerators and heat pumps is expressed in terms of the **coefficient of performance** (COP), defined as:

$$\text{COP}_R = \frac{\text{Desired output}}{\text{Required input}} = \frac{\text{Cooling Effect}}{\text{Work input}} = \frac{Q_L}{W_{\text{net,in}}}$$

$$\text{COP}_{\text{HP}} = \frac{\text{Desired output}}{\text{Required input}} = \frac{\text{Heating Effect}}{\text{Work input}} = \frac{Q_H}{W_{\text{net,in}}}$$

- The objective of a refrigerator is to remove heat (Q_L) from the cold medium;
- The objective of a heat pump is to supply heat (Q_H) to a warm medium
- COP is greater than one, meaning that we can transfer more heat than the work energy that we put in.
- $Q_H = W_{\text{in}} + Q_L$, dividing the whole by W_{in} , we get
- $\frac{Q_H}{W_{\text{net,in}}} = \frac{W_{\text{net,in}}}{W_{\text{net,in}}} + \frac{Q_L}{W_{\text{net,in}}}$
- $\text{COP}_{\text{HP}} = \text{COP}_R + 1$ for fixed values Q_L and Q_H

REFRIGERANTS

- Refrigerants are a fluid that can be easily boiled from a liquid into a vapour, and also be condensed from a vapour back into a liquid.
- Transport fluids which convey the heat energy from the low-temperature level to the high- temperature level where it gives its heat.
- Two important parameters that need to be considered in the selection of a refrigerant are the **temperatures of the two media (the refrigerated space and the environment) with which the refrigerant exchanges heat.**
- To have heat transfer at a reasonable rate, a temperature difference of **5°C to 10°C** should be maintained between the refrigerant and the medium with which it is exchanging heat.
- Some Common refrigerants and their refrigerant number
- CCl_2F_2 R-12 Freon a trade name – damage upper atmosphere (ozone) used for domestic refrigerators.
- R11 R-11 Feron used for water chiller
- NH_3 R-717 Trade name Ammonia
- Air R-729

Ammonia BP -33.34°C

R22 BP -40.8°C

34a- BP -26°C replacement for Freon and commonly used this days.

R410A- BP -48.°C

TABLE 12-1

Ozone depletion and global warming potentials of some common refrigerants.

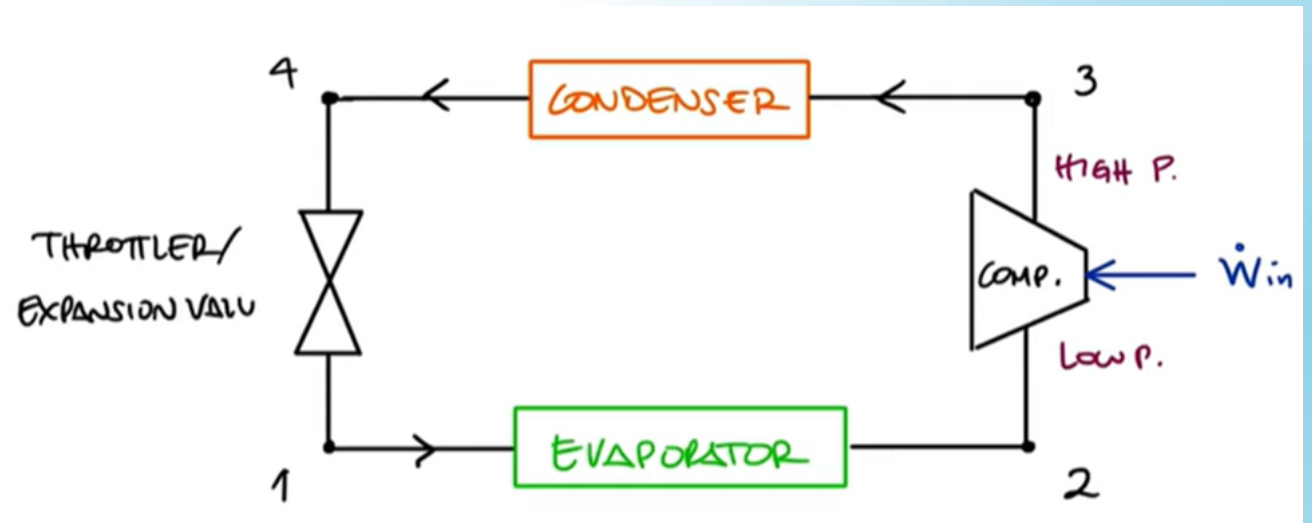
Refrigerant	Category	Ozone Depletion Potential (ODP)	Global Warming Potential (GWP)
Ammonia		0	0
R-11	CFC	1.0	4660
R-12	CFC	1.0	10,200
R-22	HCFC	0.055	1760
R-134a	HFC	0	1300
R-407C	HFC	0	1775
R-410A	HFC	0	2090
R-32	HFC	0	677

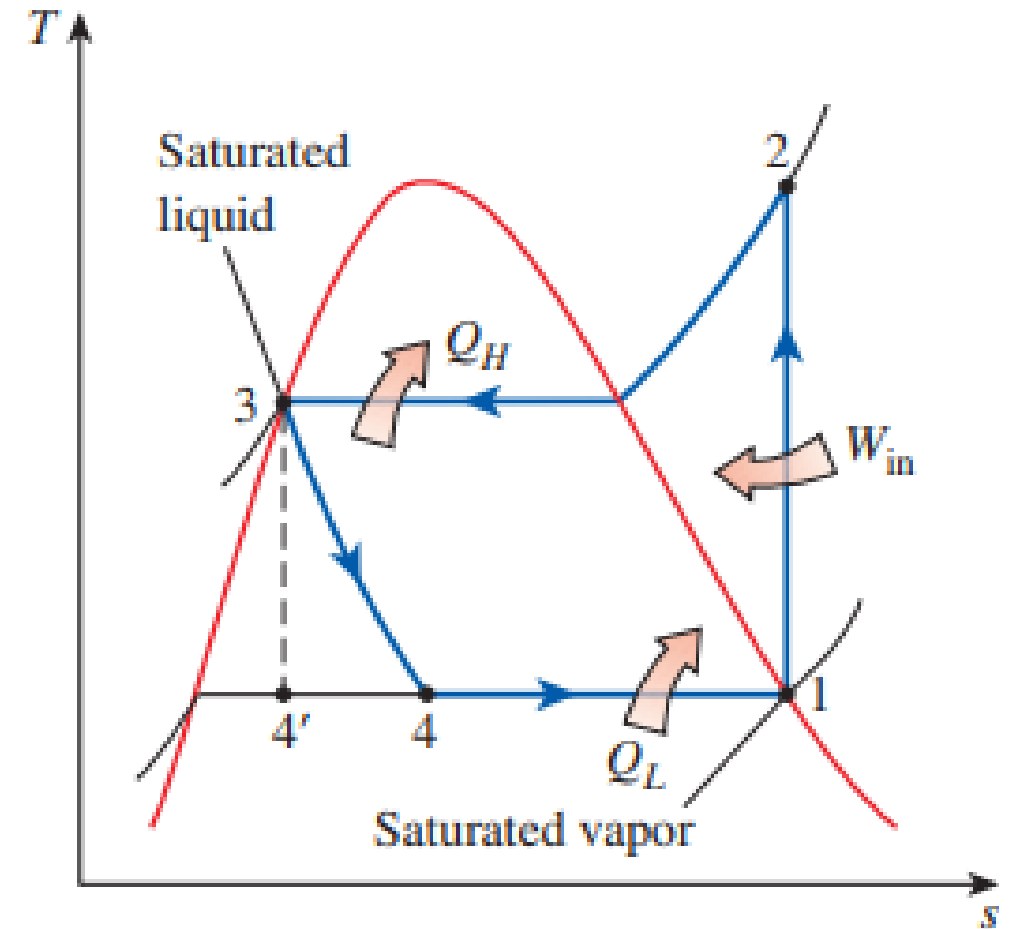
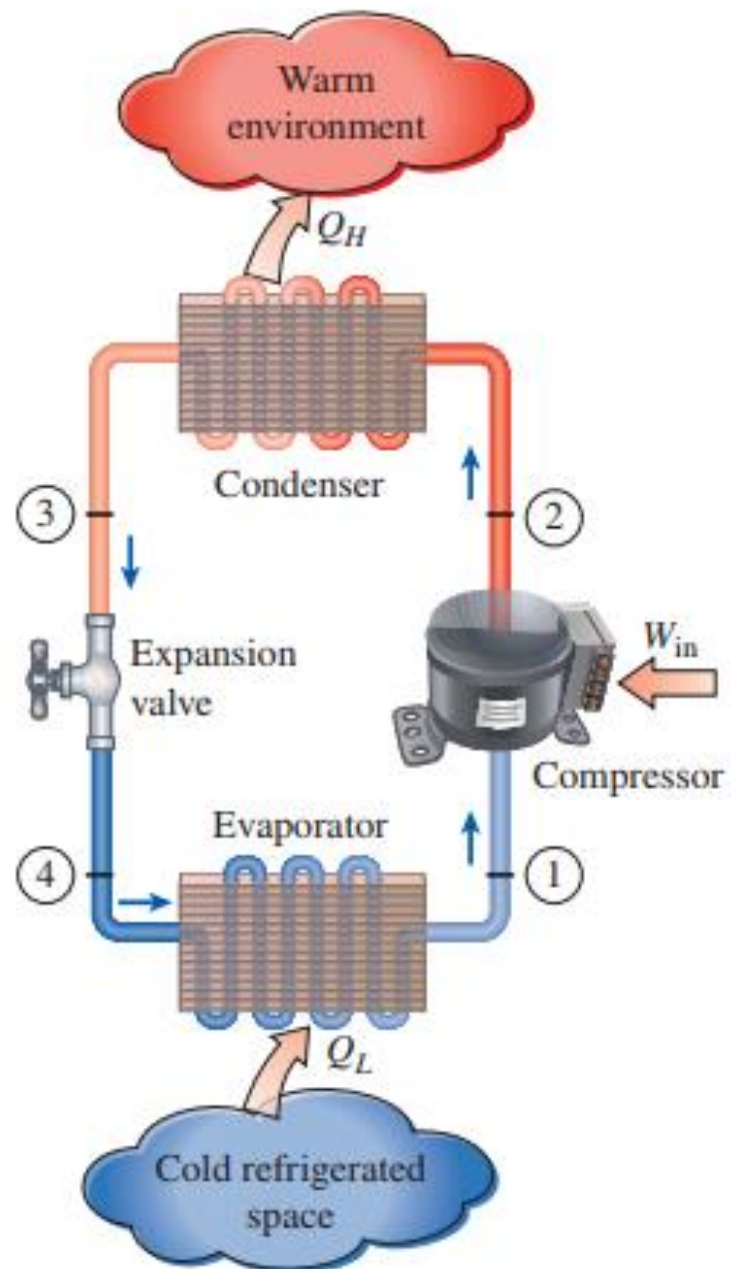
THE IDEAL VAPOR COMPRESSION REFRIGERATION CYCLE

- The **vapor compression refrigeration (VCR)** cycle is a thermodynamic process that removes heat from an enclosed space and transfers it to another.
- This is achieved by circulating a refrigerant, through four key components, where it undergoes pressure and phase changes
- The vapor-compression refrigeration cycle is the ideal model for refrigeration systems.
- This is the most widely used cycle for refrigerators, A-C systems, and heat pumps.

The four main components of the vapor compression refrigeration cycle are:

- **Compressor**
- **Condenser**
- **Expansion valve (or throttling valve)**
- **Evaporator**





Schematic and T - s diagram for the ideal vapor-compression refrigeration cycle.

- 1-2 Isentropic compression in a compressor

The refrigerant enters the compressor as saturated vapor and is compressed isentropically, increasing its pressure and temperature to become superheated vapor.

- 2-3 Constant-pressure heat rejection in a condenser

- The high-pressure superheated vapor releases heat to the surroundings in the condenser and condenses into a saturated liquid.

- 3-4 Throttling in an expansion device

- The saturated liquid passes through an expansion valve, undergoing a pressure and temperature drop that produces a cold, low-quality mixture.

- 4-1 Constant-pressure heat absorption in an evaporator

- The cold refrigerant absorbs heat from the refrigerated space in the evaporator, fully evaporates into saturated vapor, and returns to the compressor to repeat the cycle.

- The ideal vapor-compression refrigeration cycle involves an irreversible (throttling) process to make it a more realistic model for the actual systems. Replacing the expansion valve by a turbine is not practical since the added benefits cannot justify the added cost and complexity.

- **Steady-flow energy balance**

$$(q_{in} - q_{out}) + (w_{in} - w_{out}) = h_e - h_i$$

Then the COPs of refrigerators and heat pumps operating on the vapor-compression

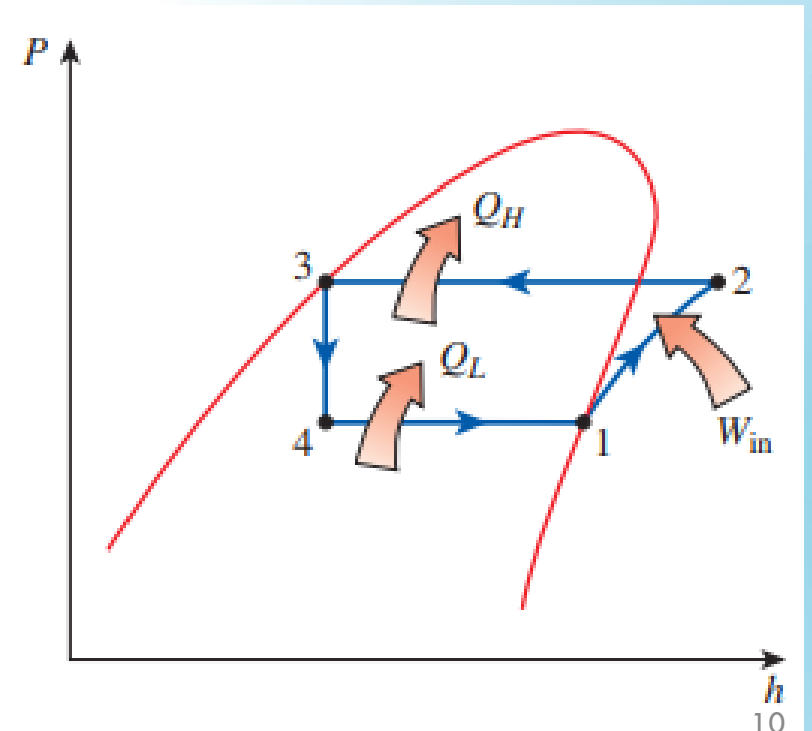
refrigeration cycle can be expressed as

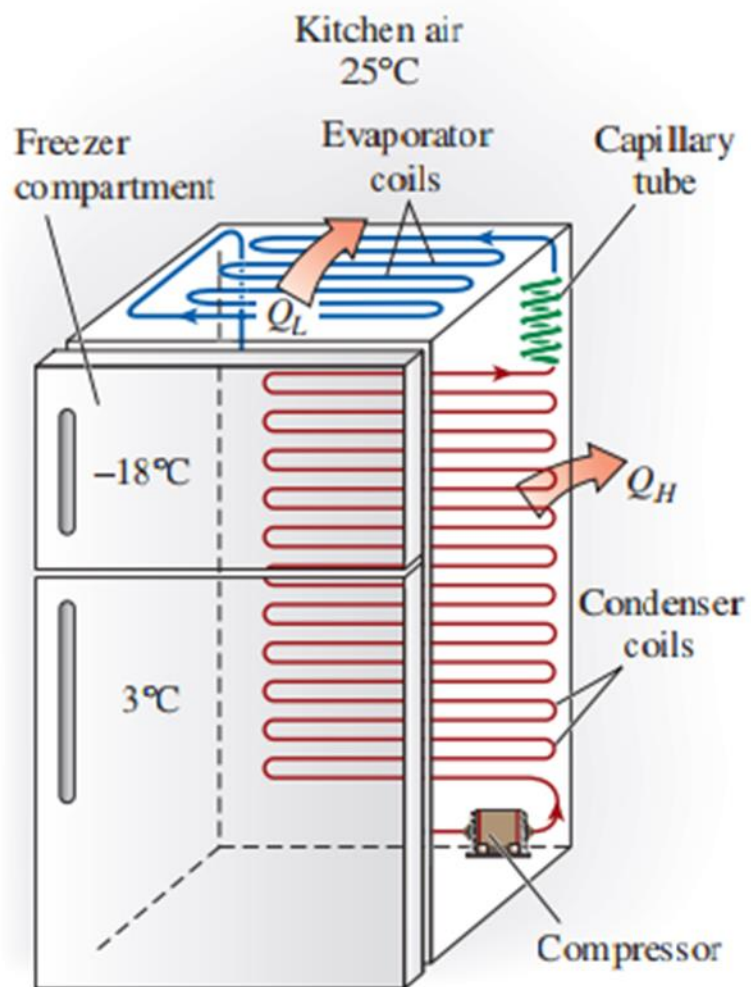
$$\text{COP}_R = \frac{q_L}{W_{\text{net,in}}} = \frac{h_1 - h_4}{h_2 - h_1}$$

and

$$\text{COP}_{\text{HP}} = \frac{q_H}{W_{\text{net,in}}} = \frac{h_2 - h_3}{h_2 - h_1}$$

Where $h_1 = h_{g@p_1}$, and $h_f@p_3$ for the ideal case.





- In a household refrigerator, the tubes in the freezer compartment where heat is absorbed by the refrigerant serve as the evaporator.
- The coils behind the refrigerator, where heat is dissipated to the kitchen air, serve as the condenser.

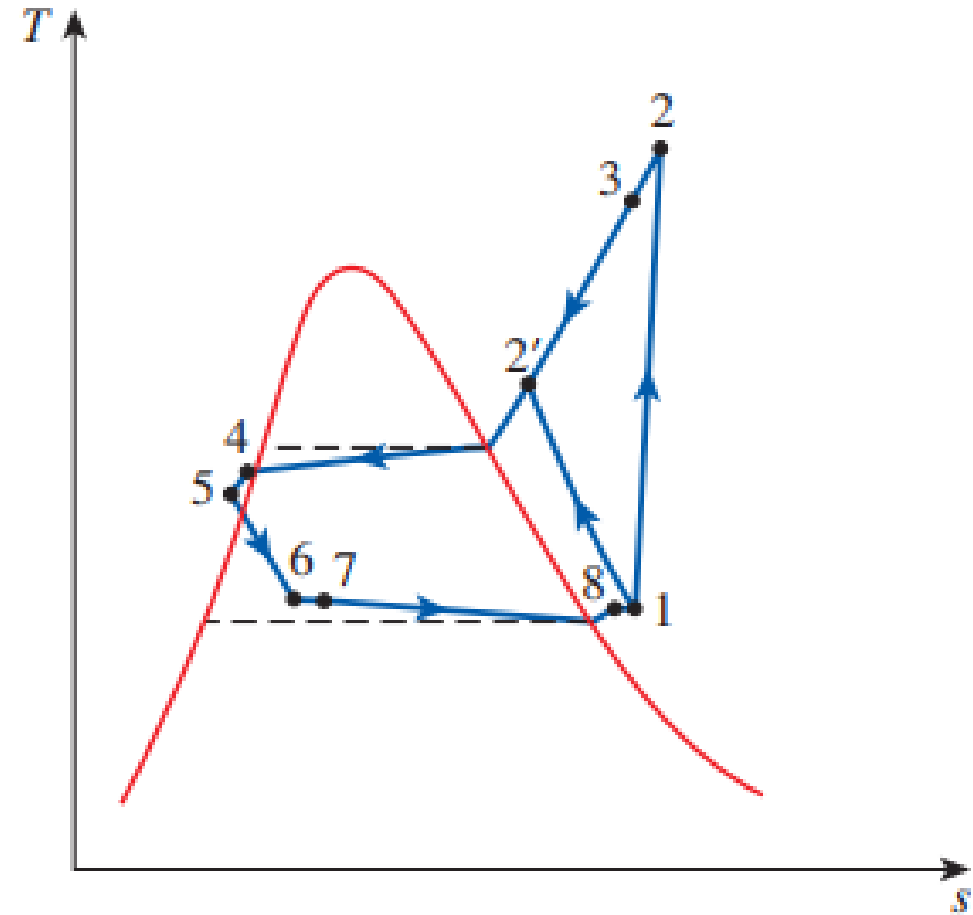
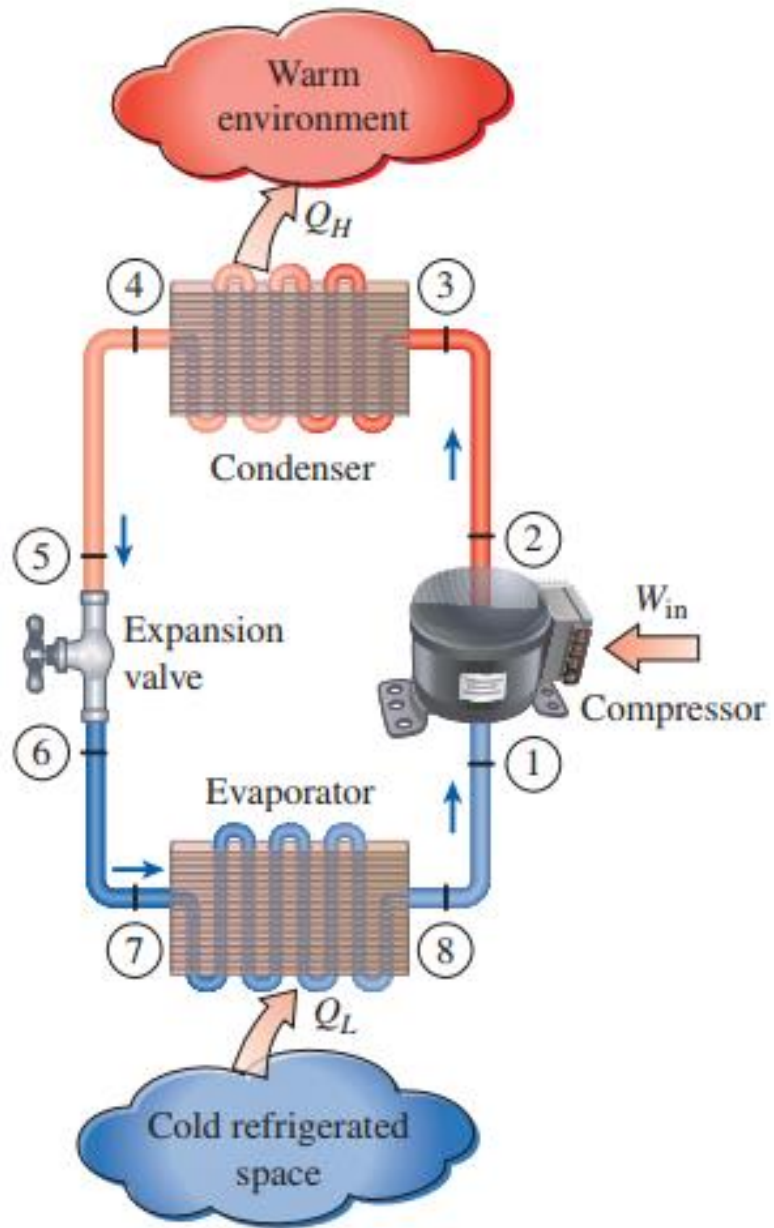
An ordinary household refrigerator.

Actual vapor-Compression Refrigeration Cycle

- An actual vapor-compression refrigeration cycle differs from the ideal one in several ways, owing mostly to the irreversibilities that occur in various components, mainly due to **fluid friction** (causes pressure drops) **and heat transfer to or from the surroundings**.
- The COP decreases as a result of irreversibilities.

DIFFERENCES

- Non-isentropic compression
- Superheated vapor at evaporator exit
- Subcooled liquid at condenser exit
- Pressure drops in condenser and evaporator



Schematic and T - s diagram for the actual vapor-compression refrigeration cycle.

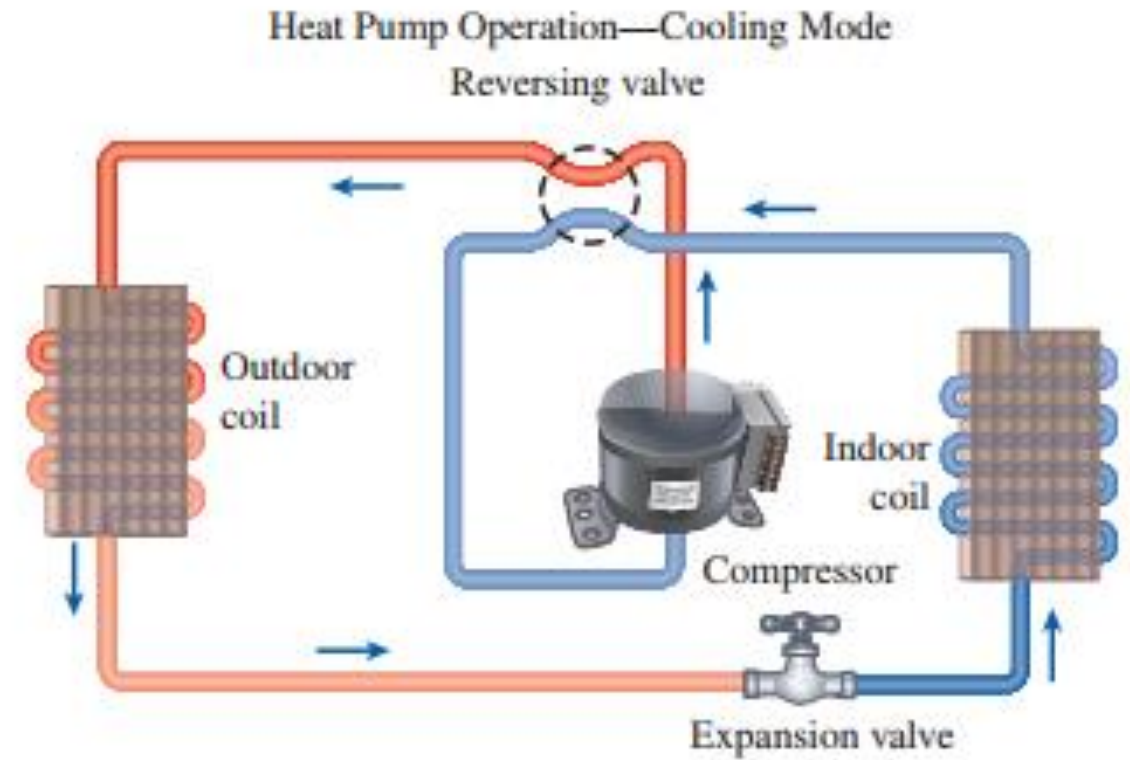
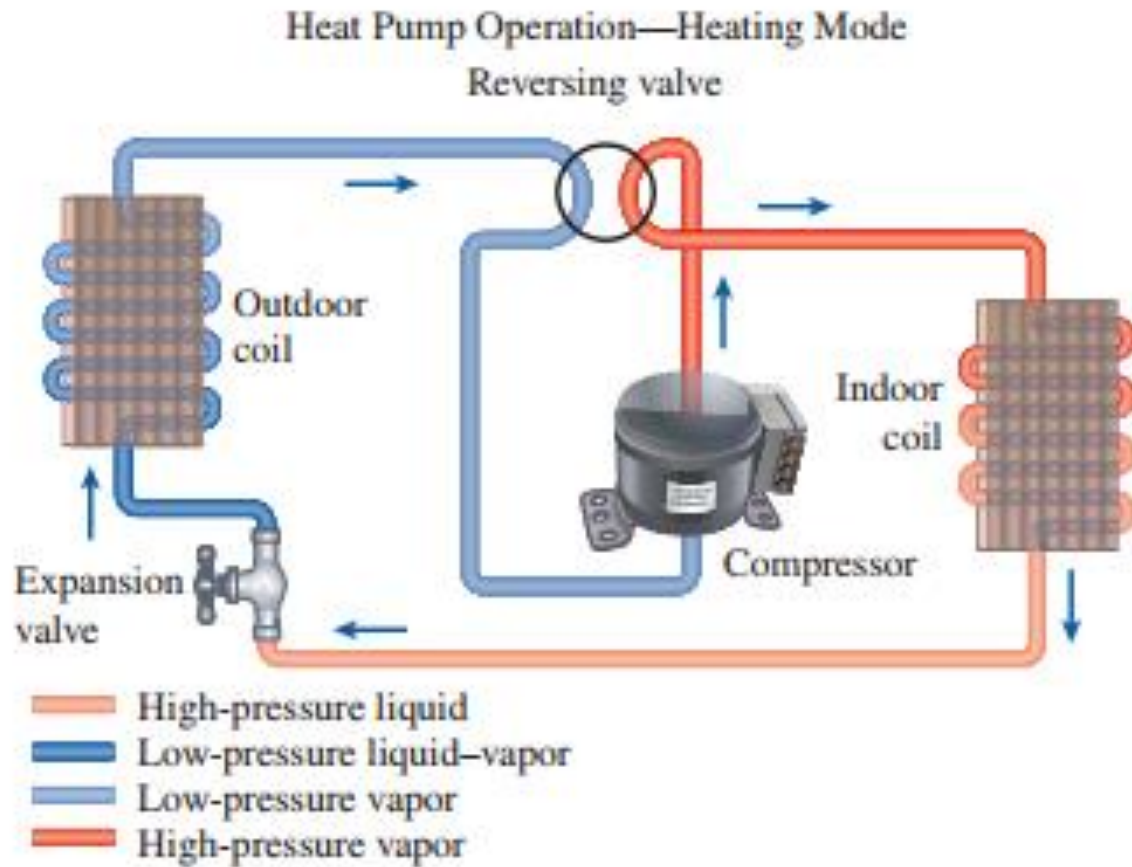
Heat Pump System

- A heat pump is a thermodynamic device that transfers heat from a low-temperature reservoir to a high-temperature reservoir by consuming external work.
- Unlike natural heat flow (from hot to cold), a heat pump reverses this direction using a working fluid and a cyclic process.

Comparison with Refrigerator:

- Both use similar cycles.
- **Heat pump:** Focuses on delivering heat to the warm space.
- **Refrigerator:** Focuses on removing heat from the cold space.
- The most common energy source for heat pumps is atmospheric air (air-to-air systems), although water and soil are used.
- The major problem with air source systems is frosting, which occurs in humid climates when the temperature falls below 2°C to 5°C .

- Water-source systems usually use well water and ground-source (geothermal) heat pumps use earth as the energy source.
- They typically have higher COPs but are more complex and more expensive to install.
- Both the capacity and the efficiency of a heat pump fall significantly at low temperatures. Therefore, most air=source heat pumps require a supplementary heating system such as electric resistance heaters or a gas furnace.
- Heat pumps are most competitive in areas that have a large cooling load during the cooling season and a relatively small heating load during the heating season. In these areas, the heat pump can meet the entire cooling and heating needs of residential or commercial buildings.



A heat pump can be used to heat a house in winter and to cool it in summer. This is accomplished by adding a reversing valve to the cycle as shown above.

- **INNOVATIVE VAPOR-COMPRESSION REFRIGERATION SYSTEMS**

The simple vapor-compression refrigeration cycle is the most widely used refrigeration cycle, and it is adequate for most refrigeration applications.

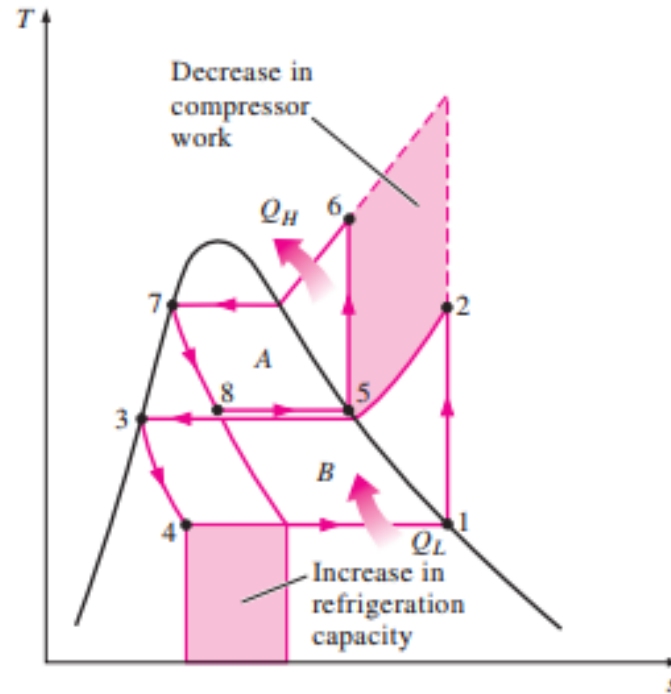
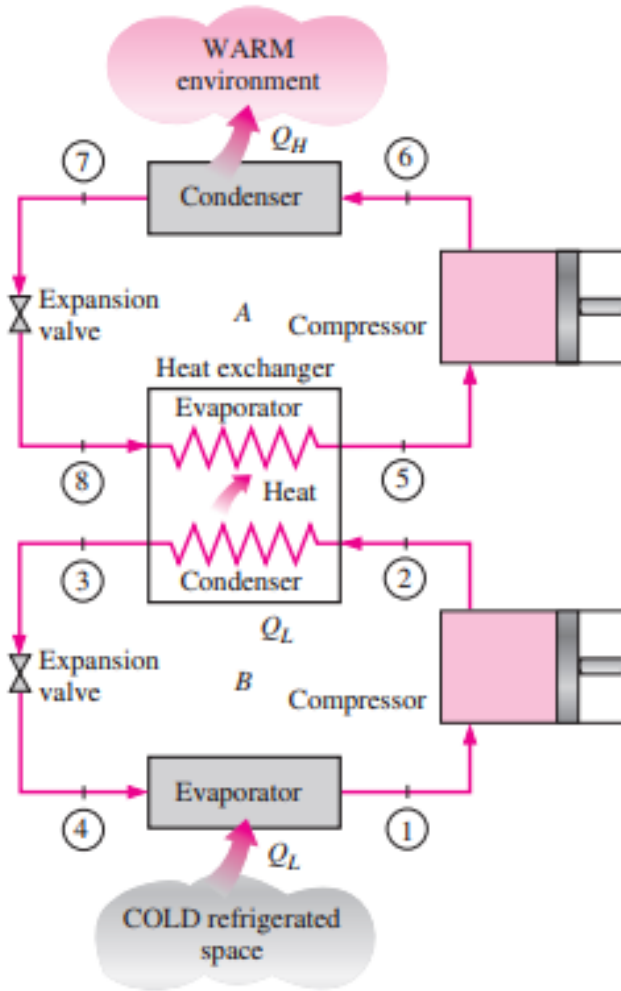
- The ordinary vapor-compression refrigeration systems are simple, inexpensive, reliable, and practically maintenance-free.
- However, for large industrial applications *efficiency*, not simplicity, is the major concern.
- Also, for some applications the simple vapor-compression refrigeration cycle is inadequate and needs to be modified.
- Some industrial applications require moderately low temperatures, and the temperature range they involve may be too large for single vapour-power cycle.
- The following cycles will be discussed:
 - **Cascade refrigeration systems**
 - **Multistage compression refrigeration systems**
 - **Multipurpose refrigeration systems with a single compressor**
 - **Liquefaction of gases**

CASCADE REFRIGERATION SYSTEMS

- Some industrial applications require moderately low temperatures ($-25\text{ }^{\circ}\text{C}$ to -75°C), and the temperature range they involve may be too large for a single vapor-compression refrigeration cycle to be practical. The solution is **cascading**.
- Cascade refrigeration is a multi-stage vapor-compression refrigeration system that employs two or more interconnected cycles, each operating with a different refrigerant optimized for a specific temperature range.
- Thermodynamically, it enhances performance by dividing the total temperature lift into smaller segments, allowing each cycle to operate closer to its ideal conditions.

$$\dot{m}_A(h_5 - h_8) = \dot{m}_B(h_2 - h_3) \longrightarrow \frac{\dot{m}_A}{\dot{m}_B} = \frac{h_2 - h_3}{h_5 - h_8}$$

$$\text{COP}_{\text{R,cascade}} = \frac{\dot{Q}_L}{\dot{W}_{\text{net,in}}} = \frac{\dot{m}_B(h_1 - h_4)}{\dot{m}_A(h_6 - h_5) + \dot{m}_B(h_2 - h_1)}$$



$$\text{COP}_{\text{R,cascade}} = \frac{\dot{Q}_L}{\dot{W}_{\text{net,in}}} = \frac{\dot{m}_B(h_1 - h_4)}{\dot{m}_A(h_6 - h_5) + \dot{m}_B(h_2 - h_1)}$$

A two-stage cascade refrigeration system with the same refrigerant in both stages.

MULTISTAGE COMPRESSION REFRIGERATION SYSTEMS

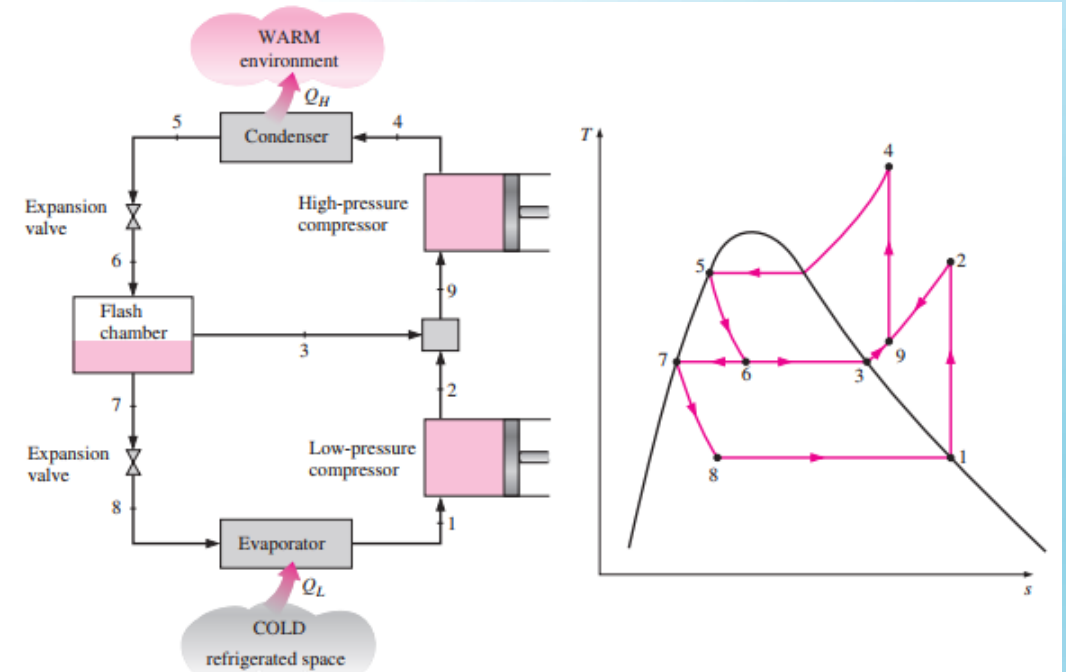
- If the temperature of the compressed refrigerant is below the environment. Mix it with refrigerant from another part of the cycle – regenerative heat exchanger called mixing chamber.
- When the fluid used throughout the cascade refrigeration system is the same, the heat exchanger between the stages can be replaced by a mixing chamber (called a flash chamber) since it has better heat-transfer characteristics. Such systems are called multistage compression refrigeration systems.

$$\sum m_e h_e = \sum m_i h_i$$

$$\sum m_9 h_9 = \sum m_3 h_3 + \sum m_2 h_2$$

$$\dot{m}_3 = x_6 \dot{m}_9 ; \quad \dot{m}_2 = (1 - x_6) \dot{m}_9$$

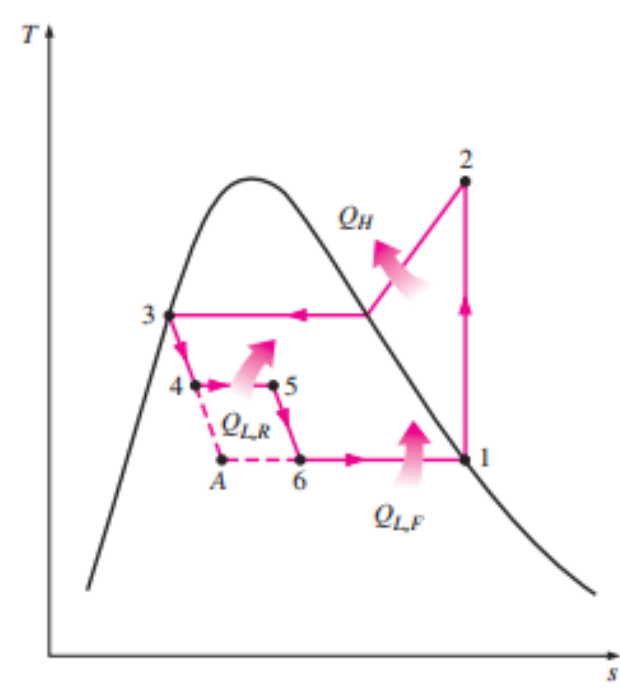
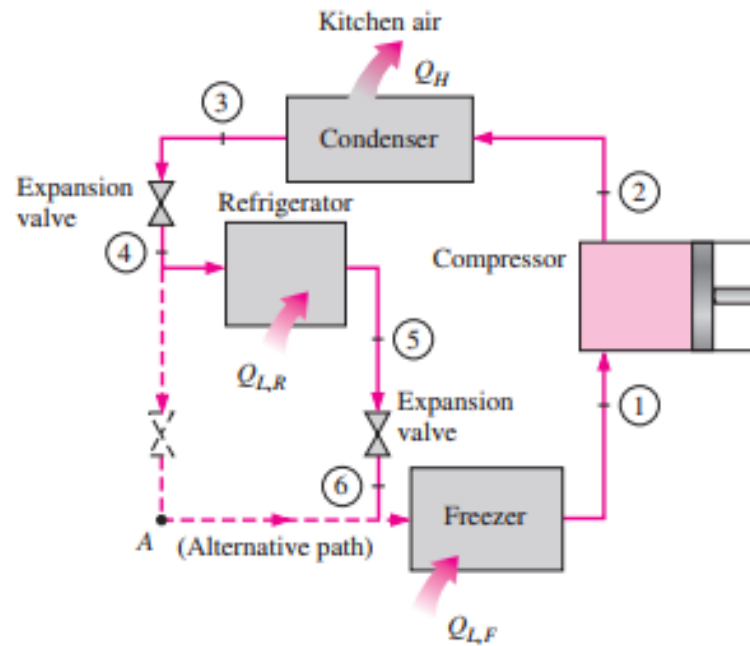
$$h_9 = x_6 h_3 + (1 - x_6) h_2$$



A two-stage compression refrigeration system with a flash chamber.

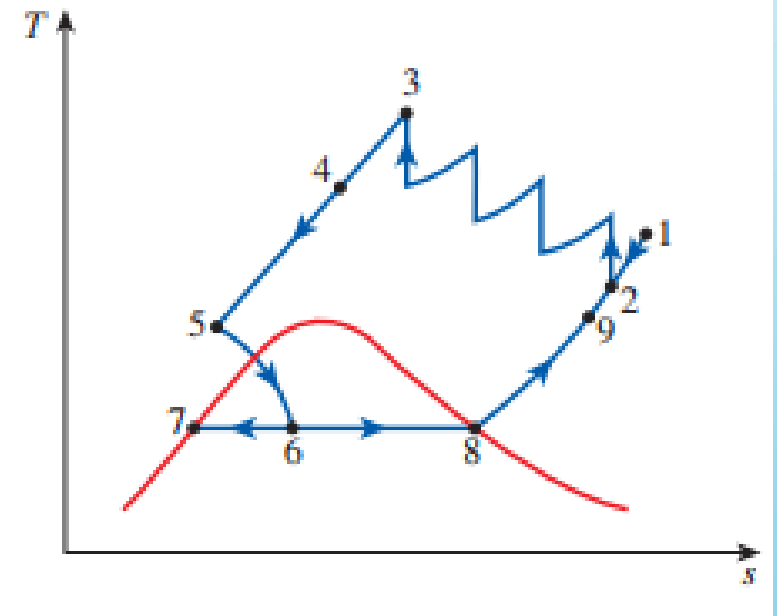
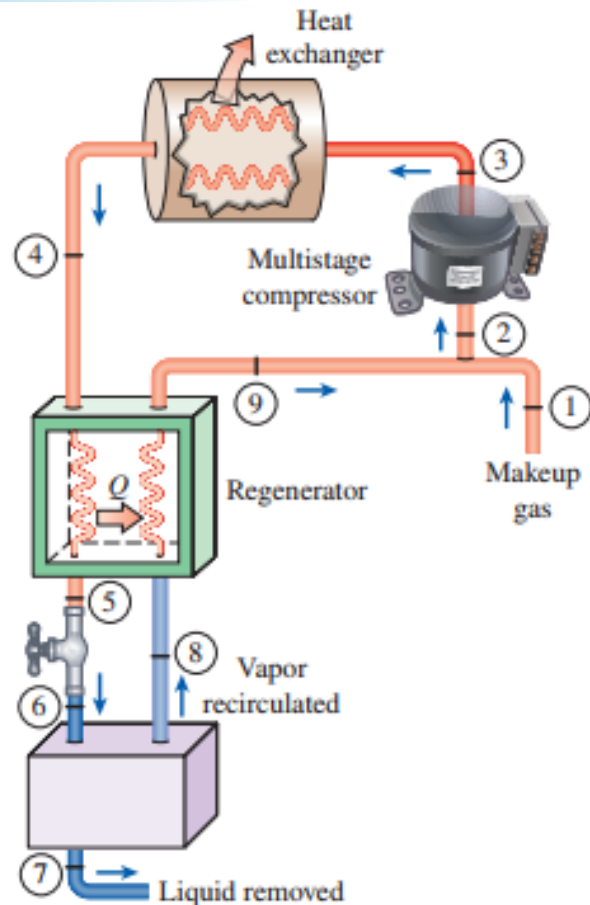
MULTIPURPOSE REFRIGERATION SYSTEMS WITH A SINGLE COMPRESSOR

- Some applications require refrigeration at more than one temperature. A practical and economical approach is to route all the exit streams from the evaporators to a single compressor and let it handle the compression process for the entire system.



LIQUEFACTION OF GASES

- Many important scientific and engineering processes at cryogenic temperatures (below about 100°C) depend on liquefied gases including the separation of oxygen and nitrogen from air, preparation of liquid propellants for rockets, the study of material properties at low temperatures, and the study of superconductivity.



Linde-Hampson system for liquefying gases.

The storage (i.e., hydrogen) and transportation of some gases (i.e., natural gas) are done after they are liquefied at very low temperatures. Several innovative cycles are used for the liquefaction of gases.

Gas Refrigeration Cycles

- A gas refrigeration system is a type of refrigeration cycle that uses a gas—typically air or helium—as the working fluid, without undergoing phase change. It operates based on the reverse Brayton (or Joule) cycle, involving compression, heat rejection, expansion, and heat absorption.
- The reversed Brayton cycle (the gas refrigeration cycle) can be used for Refrigeration.

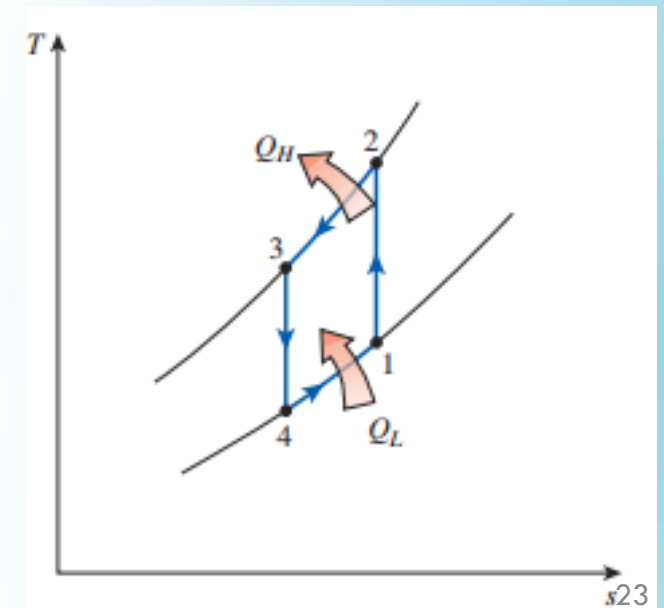
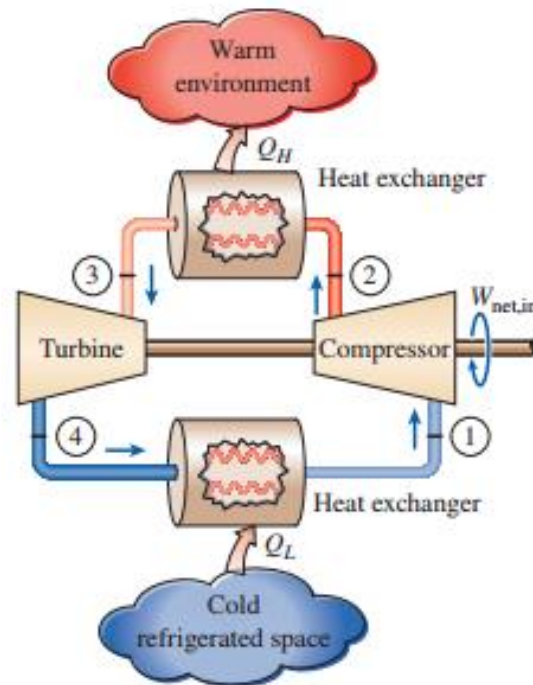
$$\text{COP}_R = \frac{q_L}{w_{\text{net,in}}} = \frac{q_L}{w_{\text{comp,in}} - w_{\text{turb,out}}}$$

$$q_L = h_1 - h_4$$

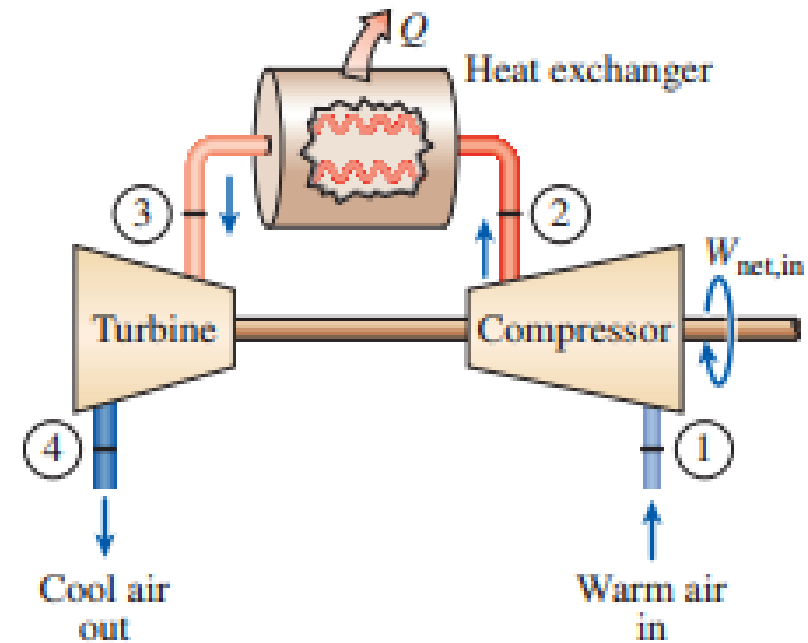
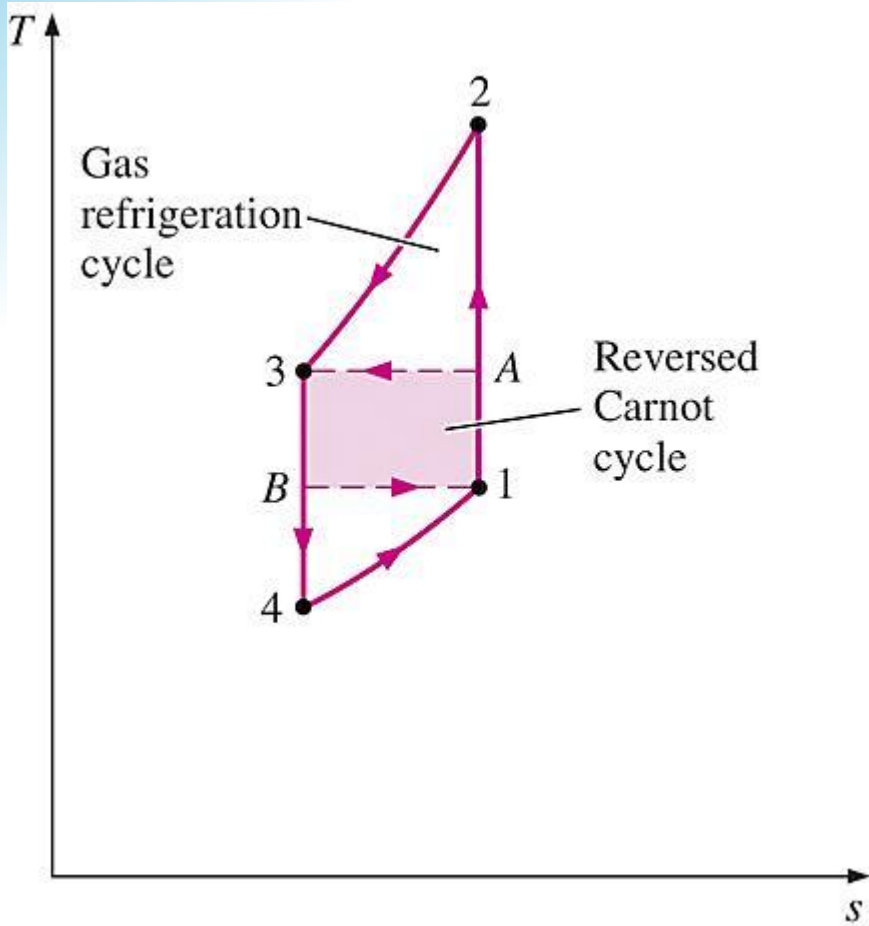
$$w_{\text{turb,out}} = h_3 - h_4$$

$$w_{\text{comp,in}} = h_2 - h_1$$

Simple gas refrigeration cycle

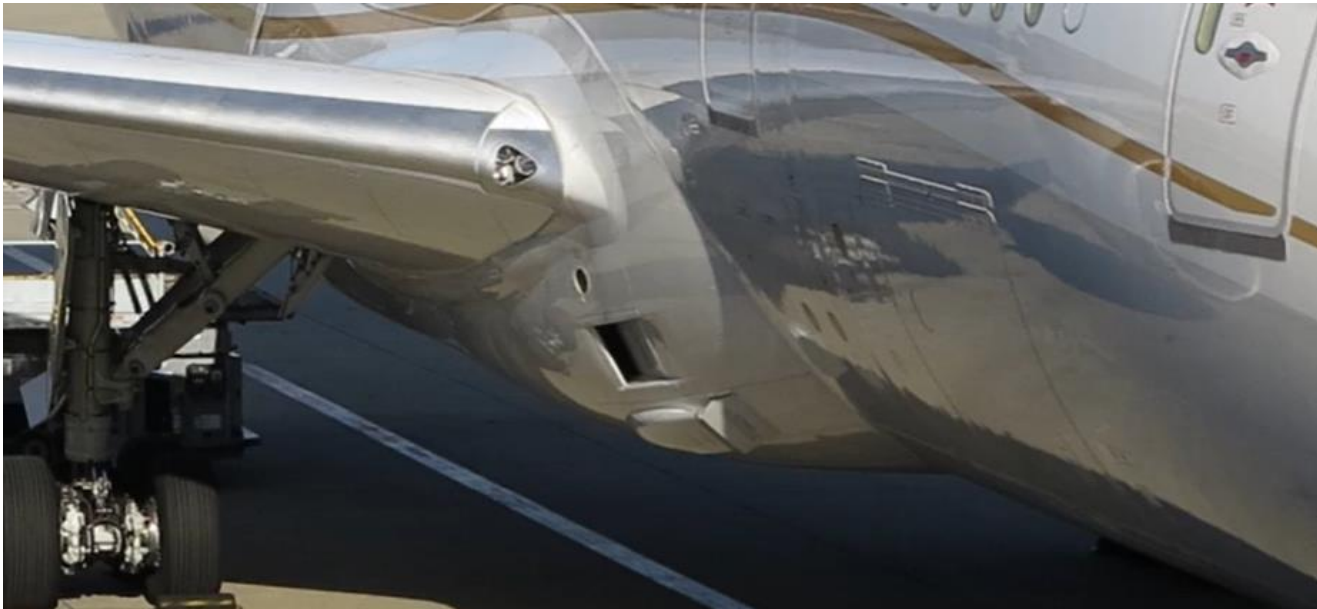


- The gas refrigeration cycles have lower COPs relative to the vapour compression refrigeration cycles or the reversed Carnot cycle.
- The reversed Carnot cycle consumes a fraction of the net work (area 1A3B) but produces a greater amount of refrigeration (triangular area under B1).

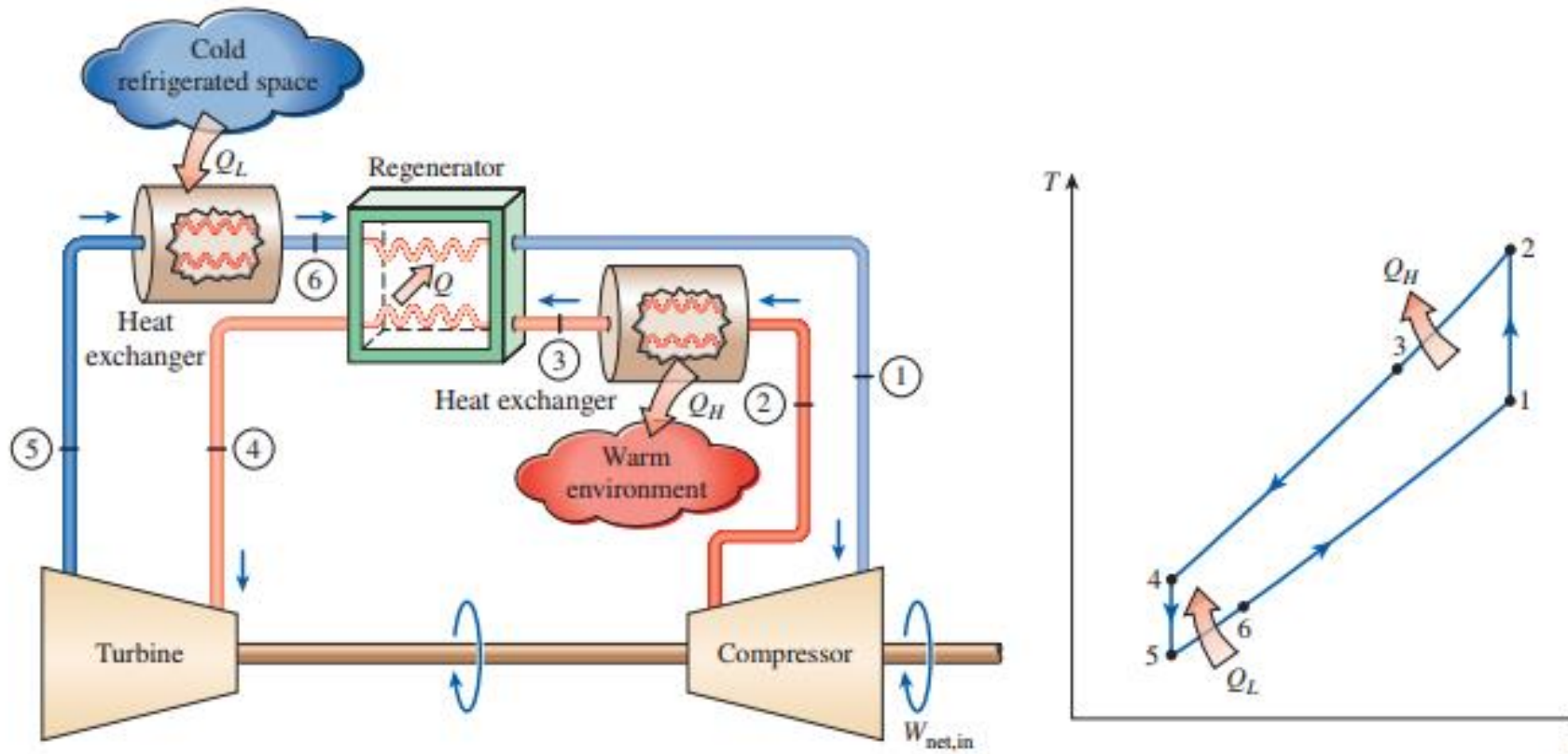


An open-cycle aircraft cooling system.

- Despite their relatively low COPs, the gas refrigeration cycles involve simple, lighter components, which make them suitable for aircraft cooling, and they can incorporate regeneration.
- Without regeneration, the lowest turbine inlet temperature is T_0 , the temperature of the surroundings or any other cooling medium.
- With regeneration, the high-pressure gas is further cooled to T_4 before expanding in the turbine.
- Lowering the turbine inlet temperature automatically lowers the turbine exit temperature, which is the minimum temperature in the cycle.



Ram air from the wing-fuselage juncture area cools compressor bleed air followed by further expansion and compression for humidity and temperature control.



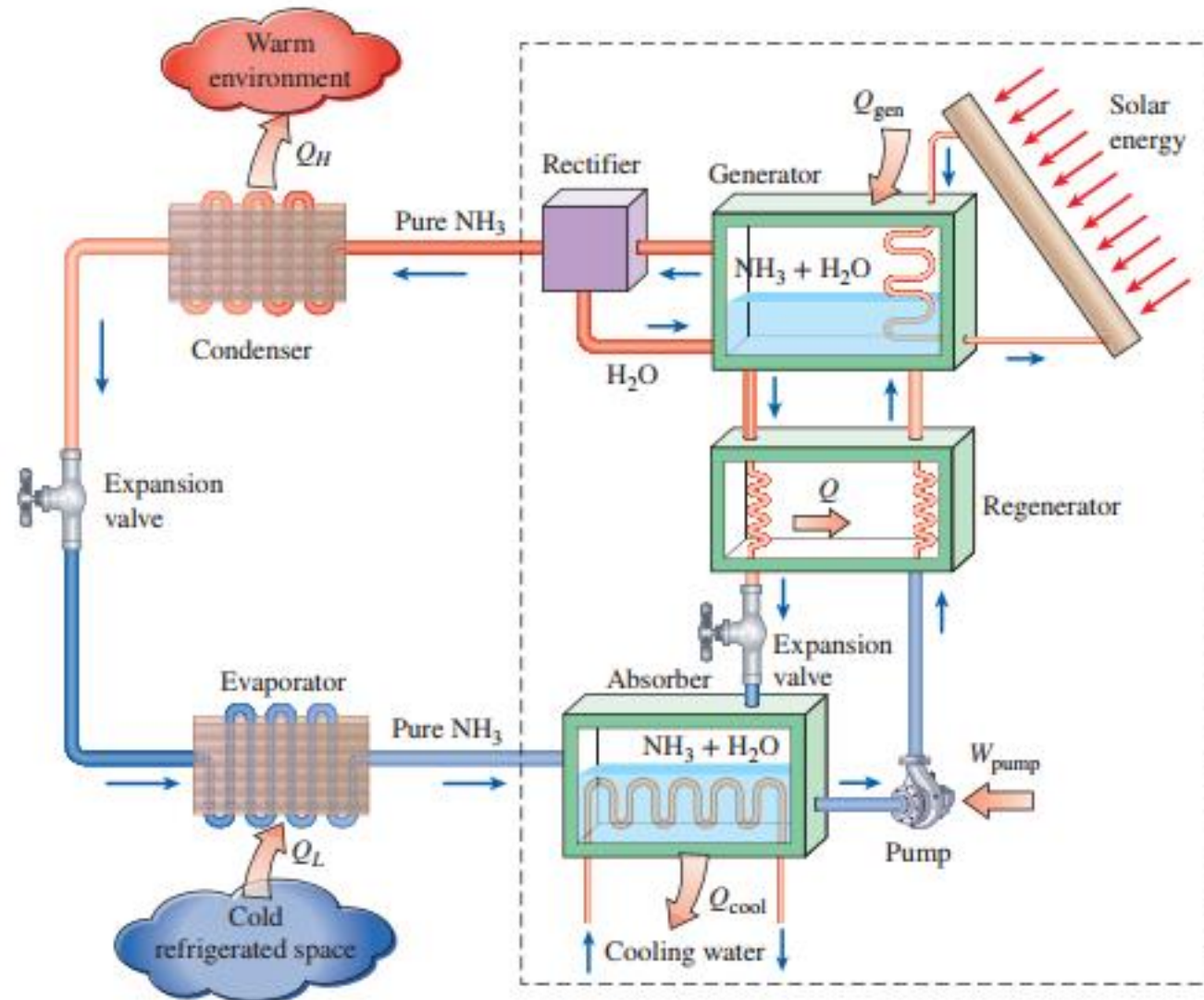
Gas refrigeration cycle with regeneration.

Extremely low temperatures can be achieved by repeating this process.

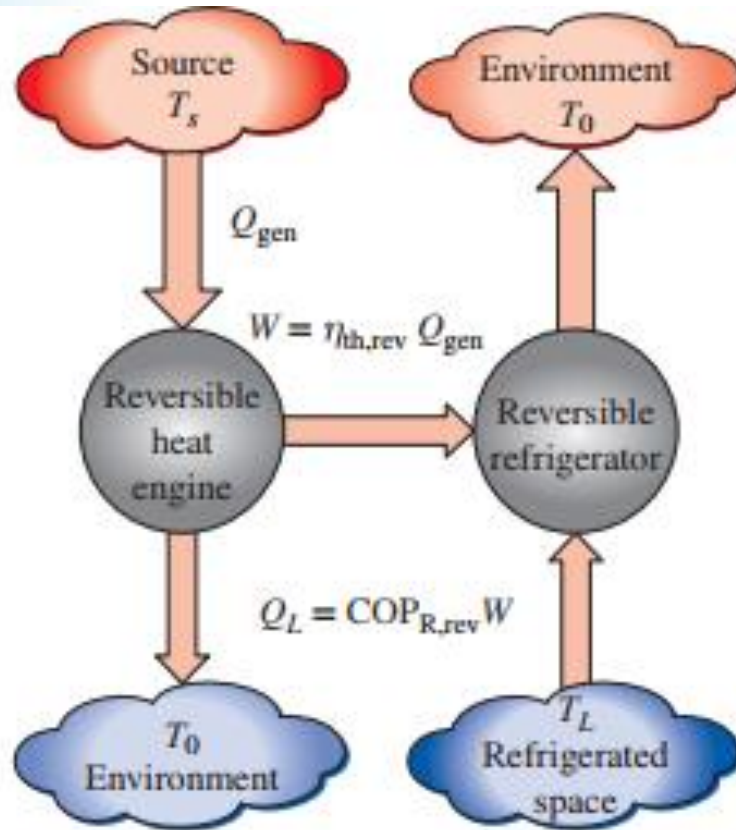
ABSORPTION REFRIGERATION SYSTEM

- Another form of refrigeration that becomes economically attractive when there is a source of inexpensive thermal energy at a temperature of 100°C to 200°C is absorption refrigeration.
- Some examples include geothermal energy, solar energy, and waste heat from cogeneration or process steam plants, and even natural gas when it is
- at a relatively low price.
- As the name implies, absorption refrigeration systems involve the absorption of a refrigerant by a transport medium.
- Other systems include water–lithium bromide and water–lithium chloride systems, where water serves as the refrigerant. These systems are limited to applications such as A-C where the minimum temperature is above the freezing point of water.

- Compared with vapor-compression systems, ARS have one major advantage: A liquid is compressed instead of a vapor and as a result the work input is very small (on the order of one percent of the heat supplied to the generator) and often neglected in the cycle analysis.
- ARS are often classified as heat-driven systems.
- ARS are much more expensive than the vapor-compression refrigeration systems. They are more complex and occupy more space, they are much less efficient thus requiring much larger cooling towers to reject the waste heat, and they are more difficult to service since they are less common.
- Therefore, ARS should be considered only when the unit cost of thermal energy is low and is projected to remain low relative to electricity.
- ARS are primarily used in large commercial and industrial installations.



Ammonia absorption refrigeration cycle.



$$W = \eta_{\text{th, rev}} Q_{\text{gen}} = \left(1 - \frac{T_0}{T_s}\right) Q_{\text{gen}}$$

$$Q_L = \text{COP}_{\text{R,rev}} W = \left(\frac{T_L}{T_0 - T_L}\right) W$$

$$\text{COP}_{\text{rev,absorption}} = \frac{Q_L}{Q_{\text{gen}}} = \left(1 - \frac{T_0}{T_s}\right) \left(\frac{T_L}{T_0 - T_L}\right)$$

The COP of absorption refrigeration systems is defined as

$$\text{COP}_{\text{absorption}} = \frac{\text{Desired output}}{\text{Required input}} = \frac{Q_L}{Q_{\text{gen}} + W_{\text{pump}}} \cong \frac{Q_L}{Q_{\text{gen}}}$$

The overall COP of an absorption refrigeration system under reversible conditions becomes

$$\text{COP}_{\text{rev,absorption}} = \frac{Q_L}{Q_{\text{gen}}} = \eta_{\text{th,rev}} \text{COP}_{\text{R,rev}} = \left(1 - \frac{T_0}{T_s}\right) \left(\frac{T_L}{T_0 - T_L}\right)$$

where T_L , T_0 , and T_s are the thermodynamic temperatures of the refrigerated space, the environment, and the heat source, respectively.

The COP of actual absorption refrigeration systems is usually less than 1.

- Air-conditioning systems based on absorption refrigeration, called absorption chillers, perform best when the heat source can supply heat at a high temperature with little temperature drop.

Examples

- 1) A refrigerator uses refrigerant-134a as the working fluid and operates on an ideal vapor-compression refrigeration cycle between 0.14 and 0.8 MPa. If the mass flow rate of the refrigerant is 0.05 kg/s, determine (a) the rate of heat removal from the refrigerated space and the power input to the compressor, (b) the rate of heat rejection to the environment, and (c) the COP of the refrigerator.
- 2) Refrigerant-134a enters the compressor of a refrigerator as superheated vapour at 0.14 MPa and 10°C at a rate of 0.05 kg/s and leaves at 0.8 MPa and 50°C. The refrigerant is cooled in the condenser to 26°C and 0.72 MPa and is throttled to 0.15 MPa. Disregarding any heat transfer and pressure drops in the connecting lines between the components, determine (a) the rate of heat removal from the refrigerated space and the power input to the compressor, (b) the isentropic efficiency of the compressor, and (c) the coefficient of performance of the refrigerator.
- 3) Consider a two-stage cascade refrigeration system operating between the pressure limits of 0.8 and 0.14 MPa. Each stage operates on an ideal vapour-compression refrigeration cycle with refrigerant-134a as the working fluid. Heat rejection from the lower cycle to the upper cycle takes place in an adiabatic counter flow heat exchanger where both streams enter at about 0.32 MPa. (In practice, the working fluid of the lower cycle is at a higher pressure and temperature in the heat exchanger for effective heat transfer.) If the mass flow rate of the refrigerant through the upper cycle is 0.05 kg/s, determine (a) the mass flow rate of the refrigerant through the lower cycle, (b) the rate of heat removal from the refrigerated space and the power input to the compressor, and (c) the coefficient of performance of this cascade refrigerator.

- 4) Consider a two-stage compression refrigeration system operating between the pressure limits of 0.8 and 0.14 MPa. The working fluid is refrigerant-134a. The refrigerant leaves the condenser as a saturated liquid and is throttled to a flash chamber operating at 0.32 MPa. Part of the refrigerant evaporates during this flashing process, and this vapor is mixed with the refrigerant leaving the low-pressure compressor. The mixture is then compressed to the condenser pressure by the high-pressure compressor. The liquid in the flash chamber is throttled to the evaporator pressure and cools the refrigerated space as it vaporizes in the evaporator. Assuming the refrigerant leaves the evaporator as a saturated vapor and both compressors are isentropic, determine (a) the fraction of the refrigerant that evaporates as it is throttled to the flash chamber, (b) the amount of heat removed from the refrigerated space and the compressor work per unit mass of refrigerant flowing through the condenser, and (c) the coefficient of performance.
- 5) An ideal gas refrigeration cycle using air as the working fluid is to maintain a refrigerated space at 223°C while rejecting heat to the surrounding medium at 27°C . If the pressure ratio of the compressor is 3, determine (a) the maximum and minimum temperatures in the cycle, (b) the coefficient of performance, and (c) the rate of refrigeration for a mass flow rate of 0.08 kg/s .

Exercises

1. A two-stage compression refrigeration system operates with refrigerant-134a between the pressure limits of 1.4 and 0.10 MPa. The refrigerant leaves the condenser as a saturated liquid and is throttled to a flash chamber operating at 0.4 MPa. The refrigerant leaving the low-pressure compressor at 0.4 MPa is also routed to the flash chamber. The vapor in the flash chamber is then compressed to the condenser pressure by the high-pressure compressor, and the liquid is throttled to the evaporator pressure. Assuming the refrigerant leaves the evaporator as saturated vapor and both compressors are isentropic, determine (a) the fraction of the refrigerant that evaporates as it is throttled to the flash chamber, (b) the rate of heat removed from the refrigerated space for a mass flow rate of 0.25 kg/s through the condenser, and (c) the coefficient of performance .
2. Repeat exercise 1 for a flash chamber pressure of 0.6 MPa.
3. Consider a two-stage cascade refrigeration system operating between the pressure limits of 0.8 and 0.14 MPa. Each stage operates on the ideal vapor-compression refrigeration cycle with refrigerant-134a as the working fluid. Heat rejection from the lower cycle to the upper cycle takes place in an adiabatic counter flow heat exchanger where both streams enter at about 0.4 MPa. If the mass flow rate of the refrigerant through the upper cycle is 0.24 kg/s, determine (a) the mass flow rate of the refrigerant through the lower cycle, (b) the rate of heat removal from the refrigerated space and the power input to the compressor, and (c) the coefficient of performance of this cascade refrigerator.

4. Consider a two-stage cascade refrigeration system operating between the pressure limits of 1.4 MPa and 160 kPa with refrigerant-134a as the working fluid. Heat rejection from the lower cycle to the upper cycle takes place in an adiabatic counterflow heat exchanger where the pressure in the upper and lower cycles are 0.4 and 0.5 MPa, respectively. In both cycles, the refrigerant is a saturated liquid at the condenser exit and a saturated vapor at the compressor inlet, and the isentropic efficiency of the compressor is 80 percent. If the mass flow rate of the refrigerant through the lower cycle is 0.11 kg/s, determine (a) the mass flow rate of the refrigerant through the upper cycle, (b) the rate of heat removal from the refrigerated space, and (c) the COP of this refrigerator.